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Nephrolithiasis

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Do.
Make.
Innovate.
Reinvent the Future.

Session Objectives

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1. Learn about the different kinds of upper tract stones we see in the US
2. Understand what factors cause these different types of stones
3. Understand what dietary factors contribute to stone formation
4. Understand what medical treatments are necessary to help prevent and treat stone disease
5. Understand when surgical intervention is needed in stone disease

Case Study

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25 yo male with no past medical history comes into the ED with a 2-week history of left testicular pain. He describes the pain as a constant, sharp, and cramp-like sensation that originates in his left groin and radiates up towards his left flank and lower back. He rates the pain as 8/10 on the pain scale and notes that it is not relieved by any over-the-counter pain medications. He denies any alleviating or exacerbating factors, and the pain is present both at rest and during physical activity. The pain has progressively increased in intensity and has been associated with intermittent episodes of nausea and vomiting. He also reports increased urinary frequency and urgency. He reports no similar episodes in the past.

On physical exam the patient appears uncomfortable due to pain. There is tenderness and guarding upon palpation of the LLQ and left flank and (+) left CVA tenderness. The rest of the PE is normal.

Case Study

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Based on the history and physical exam what differentials do we want to think of for this patient?

Case Study

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Differentials:

- Nephrolithiasis
- Pyelonephritis
- UTI
- STI
- Acute intestinal obstruction
- Diverticulitis
- Herpes zoster

Kidney Stone Etiology

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- 80% of kidney stones are calcium-based stones
- Male to female ratio = 3:1
- Occur when normally soluble material becomes supersaturated and begins the process of crystal formation
- Types of kidney stones:
  1. Calcium
  2. Struvite
  3. Uric Acid
  4. Cystine

Calcium Stones

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Calcium Oxalate

- More common
- Can be caused by:
  - Ethylene glycol (antifreeze) ingestion
  - Vit C overuse
  - Hypocitraturia
  - LOW urine pH
  - Malabsorption
- Radiopaque on x-ray
- Envelope or dumbbell shaped

Calcium Phosphate

- Can be caused by:
  - HIGH urine pH
- Radiopaque on x-ray
- Hyperdense on CT
- Wedge-shaped prism crystals

Struvite Stones

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- Ammonium magnesium phosphate
- Staghorn caliculi
- 15% of stones
- Caused by:
  - Infection with urease + bugs (e.g. *Proteus mirbilis, Staphylococcus saprophyticus, Klebsiella)* --> hydrolyze urea to ammonia and cause urine alkalinization (increased urine pH)
- Radiopaque on x-ray
- Hyperdense on CT
- Coffin lid shaped crystals

Uric Acid Stones

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- Uric acid → end product of purine metabolism and excreted in urine
- 5% of all stones
- Caused by:
  - LOW urine pH (below 5.5)
  - Decreased urine volume
  - Arid climates
  - Hyperuricosuria
- Strong association with:
  - Hyperuricemia (e.g. gout)
  - Diseases with increased cell turnover (e.g. leukemia)
- Radiolucent on x-ray
- Rhomboid or rosette shaped crystals

Cystine Stones

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- Caused by a hereditary condition in which **C**ystine-reabsorbing PCT transport loses function → cystinuria
  - Poor reabsorption of certain amino acids → **O**rnithine, **L**ysine, **A**rginine (**COLA**)
- Sodium cyanide nitroprusside test is POSITIVE
- Faintly/partially radiopaque on x-ray
- Moderately radio-dense on CT
- Hexagonal shaped crystals

Major Risk Factors

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INTRINSIC

- Urinary stasis/obstruction
- Urine composition
- Genetics
- Disease state
  - RTA type 1
  - Hyperparathyroidism
  - UTI
  - Medullary Sponge Kidney
  - Sarcoidosis
- Metabolic

EXTRINSIC

- Water intake
  - Decreased volume of fluid intake
- Diet
  - Ingestion of purines, oxalates, calcium, phosphate, and sugar sweetened beverages
- Medications
  - Carbonic anhydrase inhibitors
  - Long term glucocorticoids
  - Laxative
  - Loop diuretics

Major risk factors for calcium stones

Urinary
Lower volume
Higher calcium
Higher oxalate (CaOx stones)
Lower citrate
Higher pH (CaP stones)
Anatomic
Medullary sponge kidney
Horseshoe kidney
Diet
Lower fluid intake
Lower dietary calcium
Higher oxalate
Lower potassium
Higher sodium
Higher sucrose
Higher fructose
Lower phytate
Higher vitamin C
Other medical conditions
Primary hyperparathyroidism
Gout
Obesity
Diabetes mellitus
Distal renal tubular acidosis
Inflammatory bowel disease
Malabsorptive bariatric surgery
Short bowel syndrome

CaOx: calcium oxalate; CaP: calcium phosphate.

Risk Factors for Nephrolithiasis

Non-Dietary

- **Family History: The chances of developing kidney stones are almost 2.5 times greater in these individuals**
- **Anatomic Abnormalities: horseshoe-shaped kidneys, polycystic kidneys, etc**
- **Systemic Conditions like hyperparathyroidism and Diabetes mellitus**
- **Increased Body Mass Index (BMI)**
- **History of Gout**
- **Environmental: Individuals living and working in a hot climate will have concentrated urine**

iClinic
The Virtual Hospital

Dietary

- **Low fluid intake.**
- **High consumption of animal protein.**
- **An increased amount of sodium in diet.**
- **A diet rich in oxalates (salt of oxalic acid)**
- **Decreased calcium intake.**
- **Excess consumption of refined sugars**

Urinary

- **Hypercalciuria: increased concentration of calcium in the urine**
- **Hyperoxaluria: increased concentration of calcium oxalate crystals in the urine**
- **Hyperuricosuria: the presence of an excess amount of uric acid in the blood**
- **Hypocitraturia: reduced concentration of citrate in the urine**
- **Abnormal urine pH**

Clinical Manifestations

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ASYMPTOMATIC

- Incidental finding on imaging for other purpose

PAIN

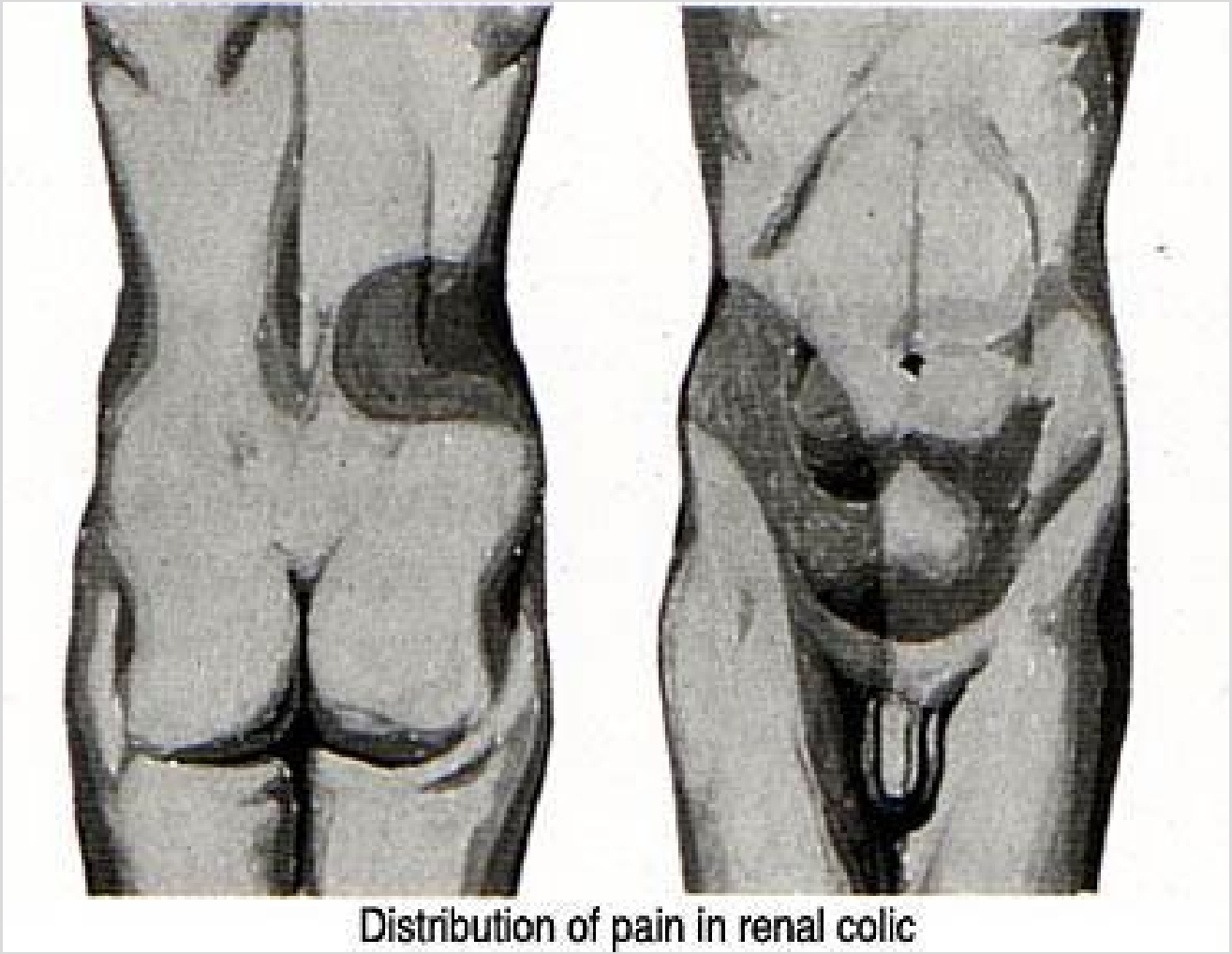
- Unilateral flank pain
- Colicky pain radiating to groin

HEMATURIA

- Gross or microscopic hematuria occurs in the majority of symptomatic kidney stones

OTHER

- Nausea and vomiting, dysuria, and urinary urgency



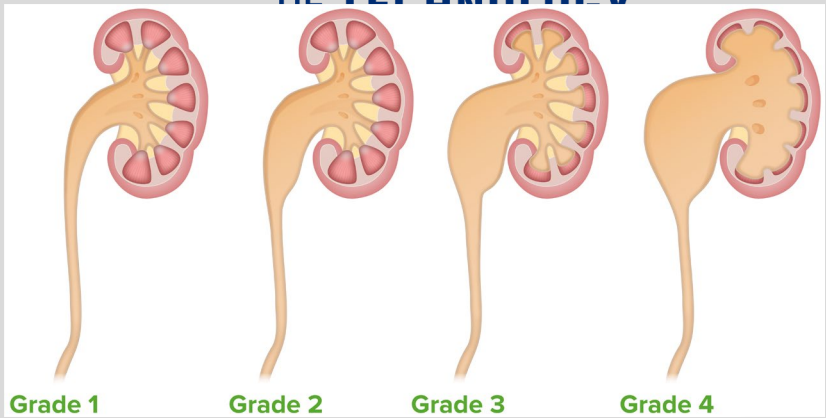
Distribution of pain in renal colic

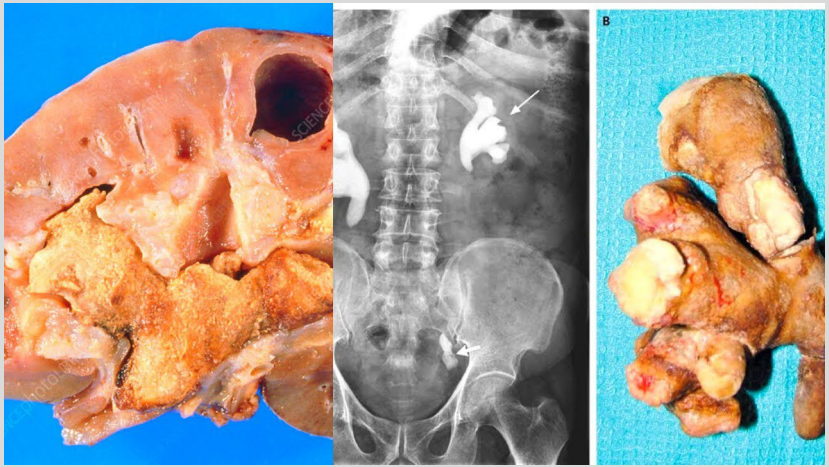
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Complications

- Prolonged urinary obstruction by a stone → hydronephrosis
- Infection proximal to the stone = emergency
- Staghorn caliculi → kidney failure if present B/L

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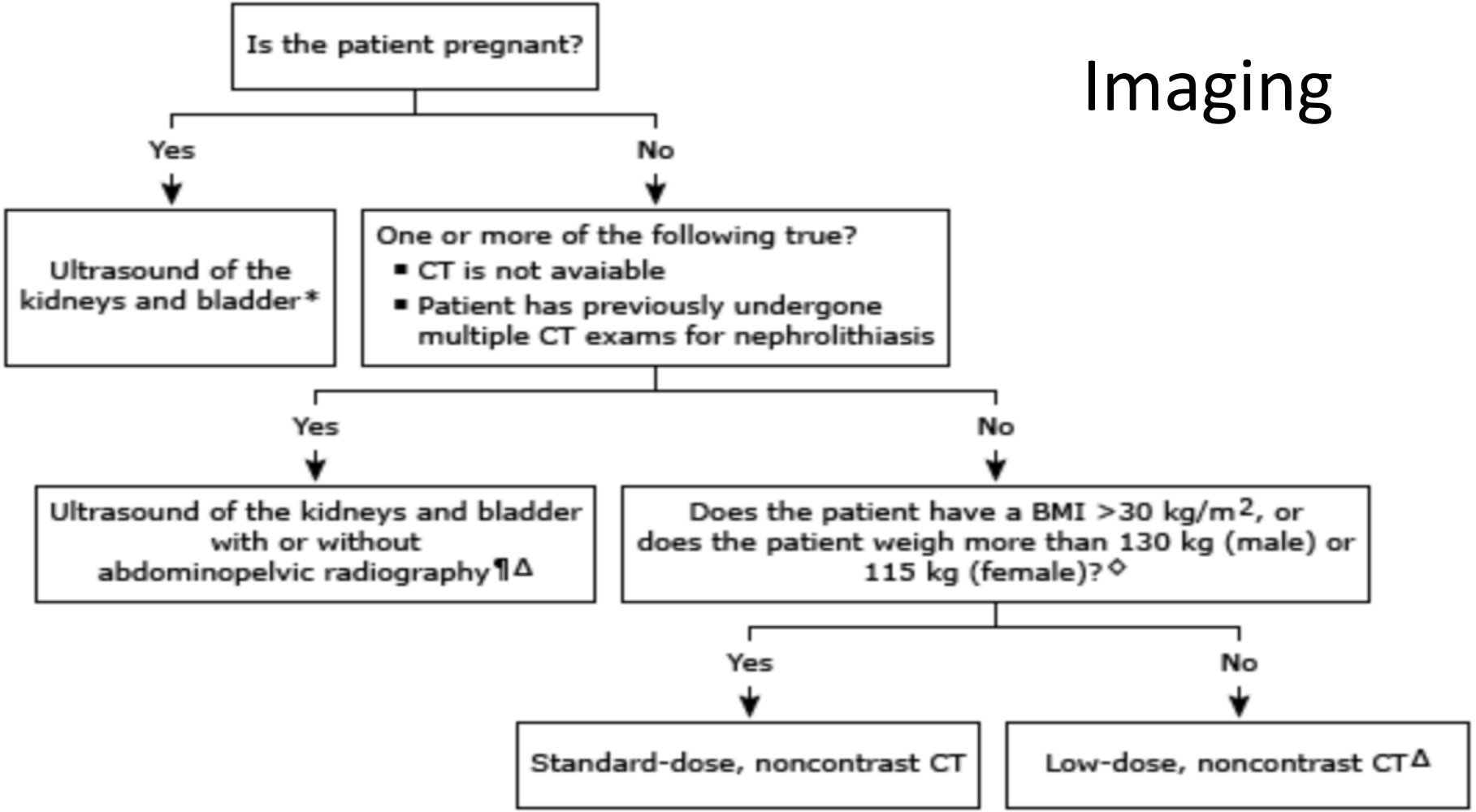


Work-up

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- Labs
  - BMP → kidney function
  - Urinalysis → infection and hematuria; urine pH
- Imaging (see chart on next slide)

Imaging


Is the patient pregnant?
Yes
No
Ultrasound of the kidneys and bladder*
One or more of the following true?
CT is not avaiable
Patient has previously undergone multiple CT exams for nephrolithiasis
Yes
No
Ultrasound of the kidneys and bladder with or without abdominopelvic radiography¶Δ
Does the patient have a BMI >30 kg/m2, or does the patient weigh more than 130 kg (male) or 115 kg (female)?◇
Yes
No
Standard-dose, noncontrast CT
Low-dose, noncontrast CTΔ

Imaging Clues for Kidney Stones

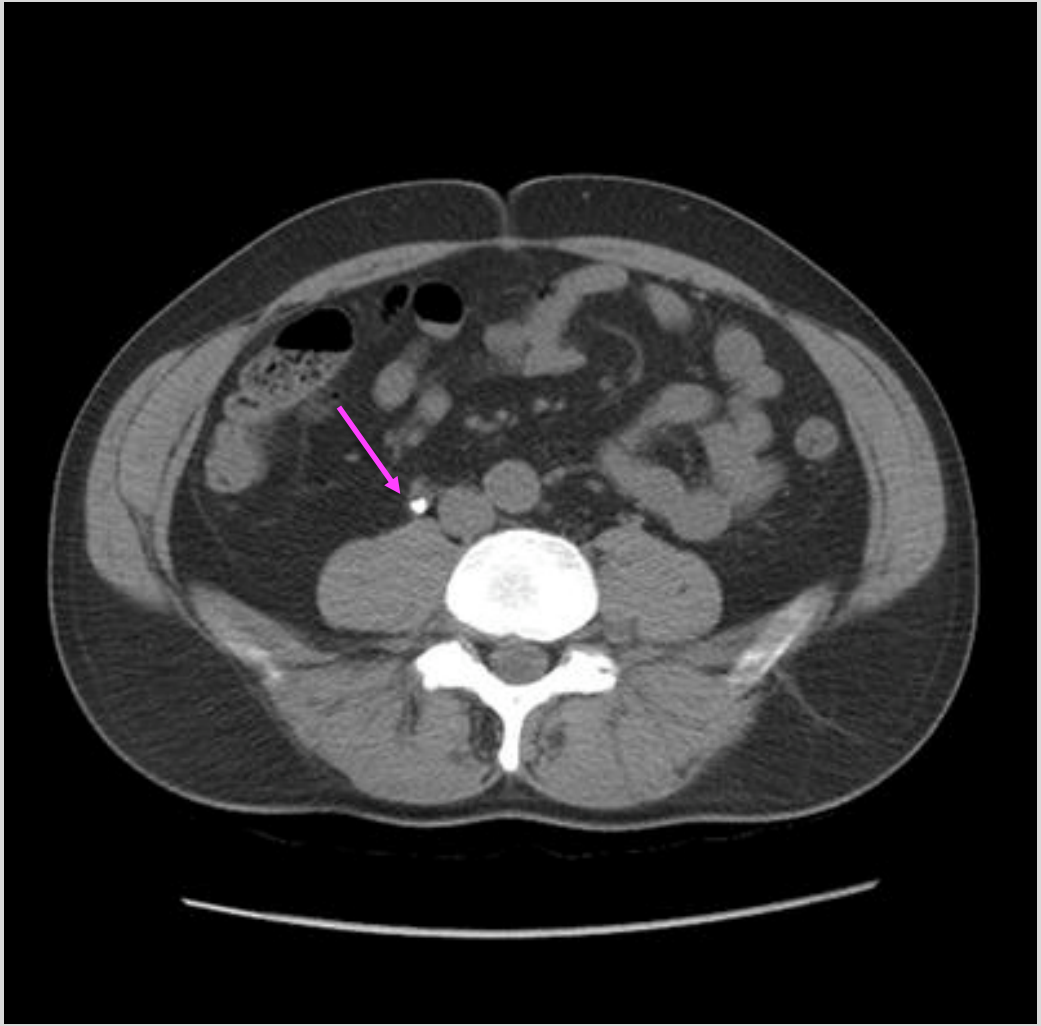
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- Calcium, struvite, and cystine stones are radiopaque (seen as white on plain films)
- Uric acid stones are radiolucent (cannot be seen on an x-ray)
- Radiolucent or undetected stones can be from
    a. Protease inhibitors (indinavir, atazanavir)
    b. Xanthine
    c. Dihydroxyadenine
- Large stones in the renal pelvis are usually struvite
- Nephrocalcinosis (excess calcium deposition in kidneys) is usually associated with calcium phosphate, less so calcium oxalate
- Medullary sponge kidney: associated with calcium stones



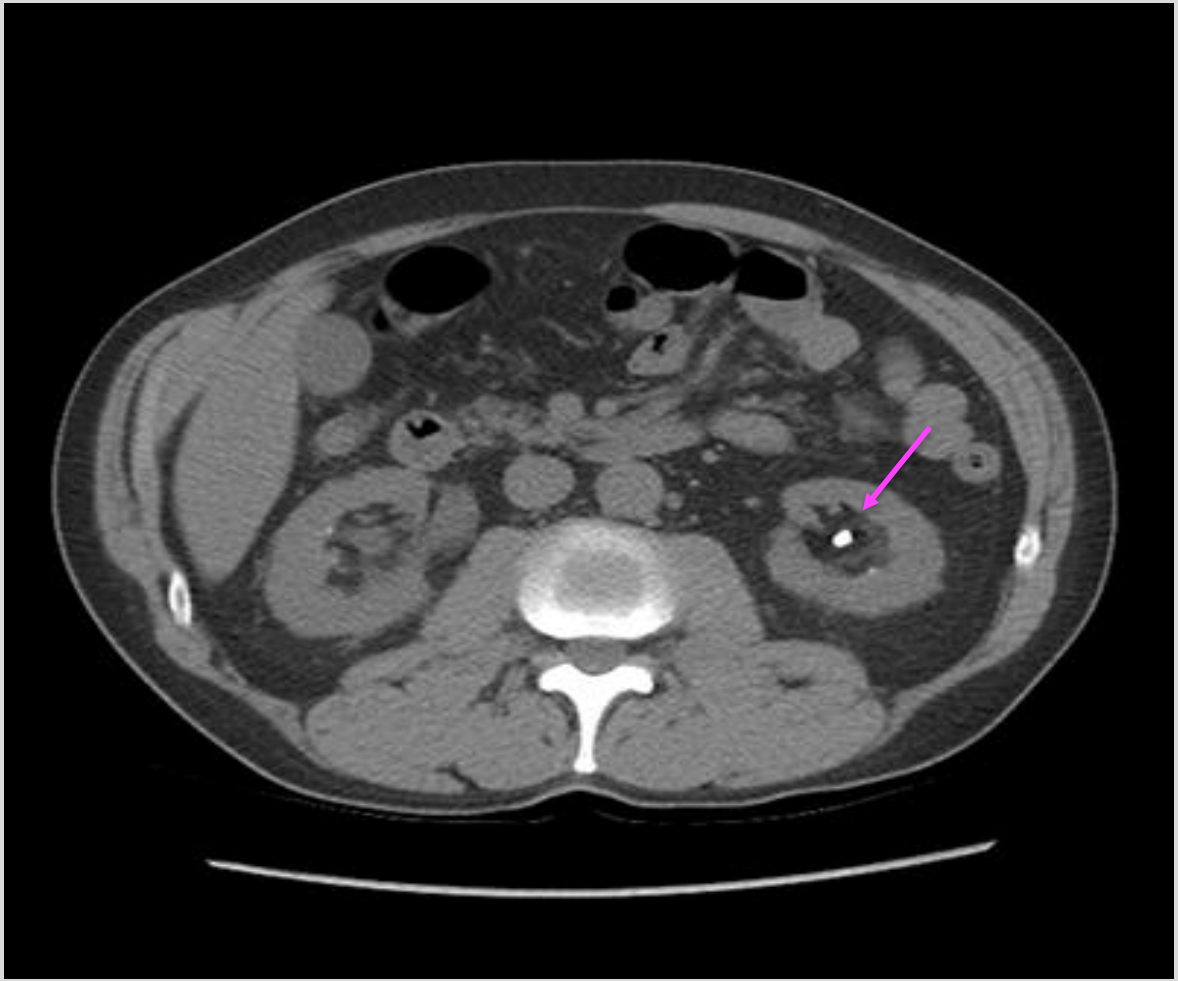
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KUB



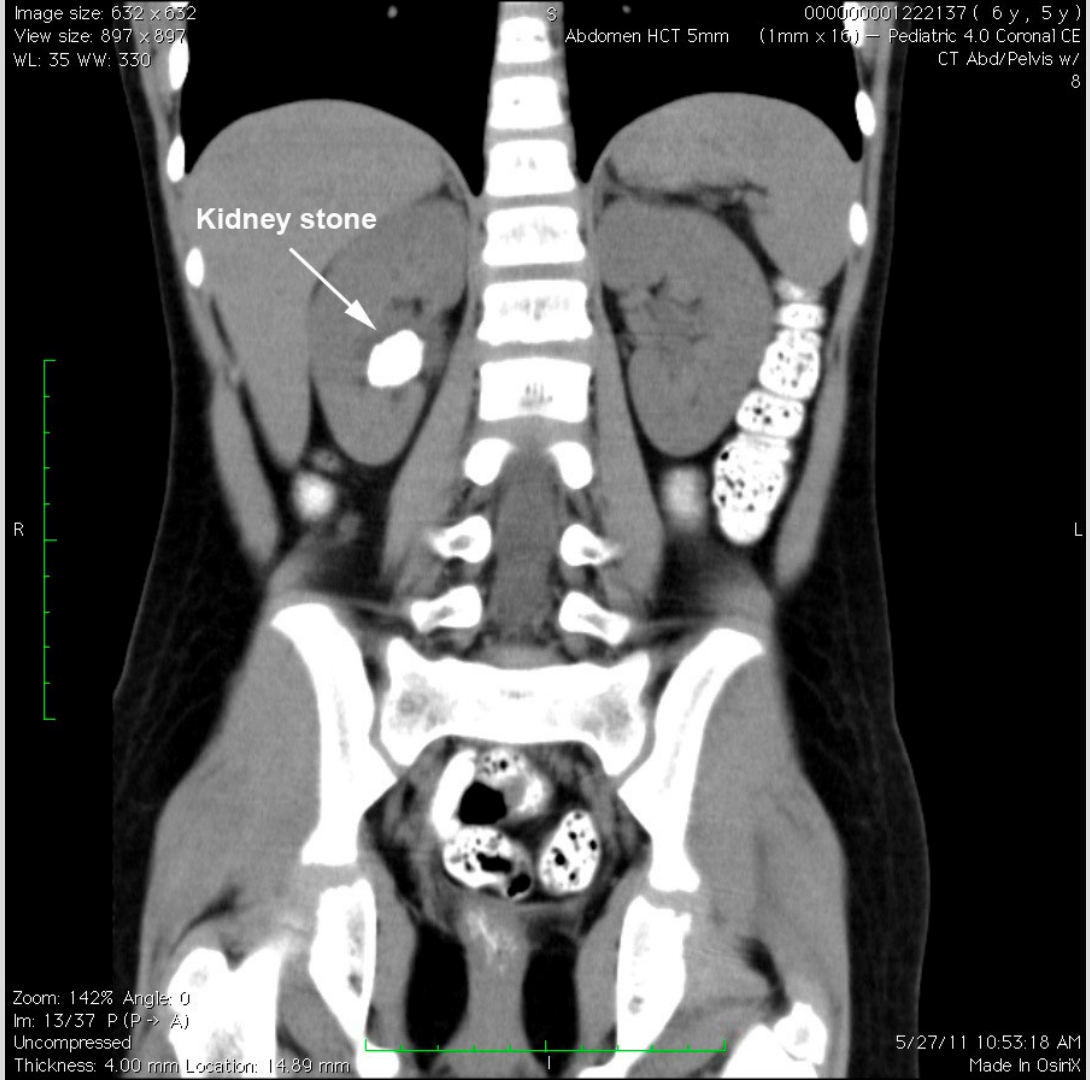
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Ureteral Stone



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Kidney Stone



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Staghorn Calculus



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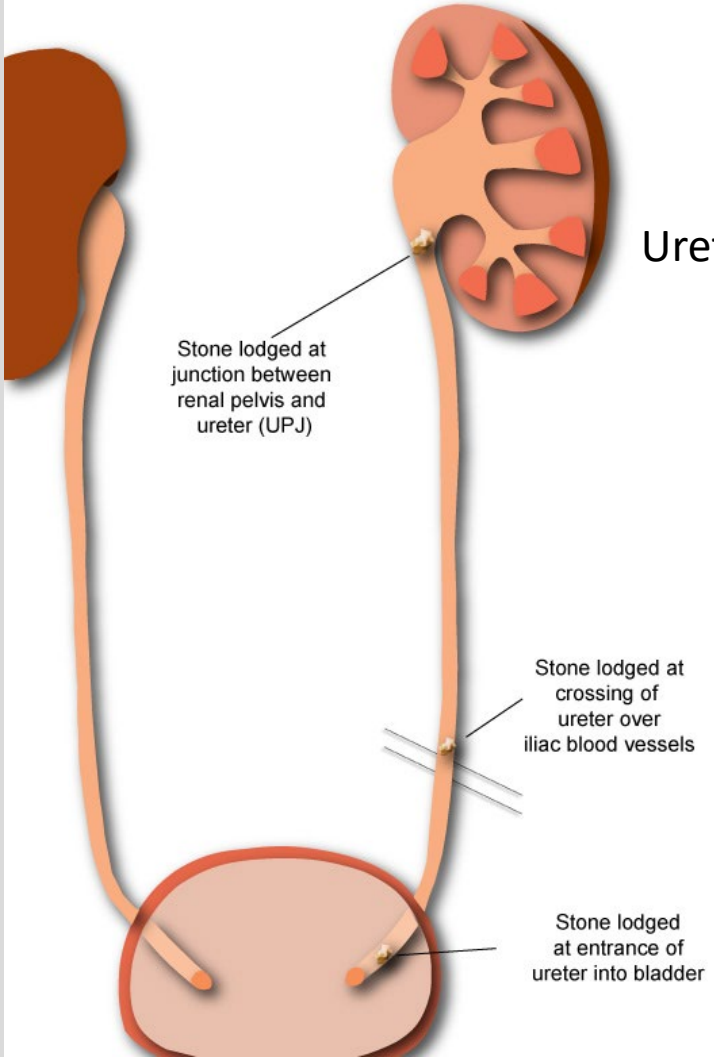
Ureter Obstruction



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Ultrasound

Where do kidney stones commonly cause obstruction?



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Ureteropelvic junction

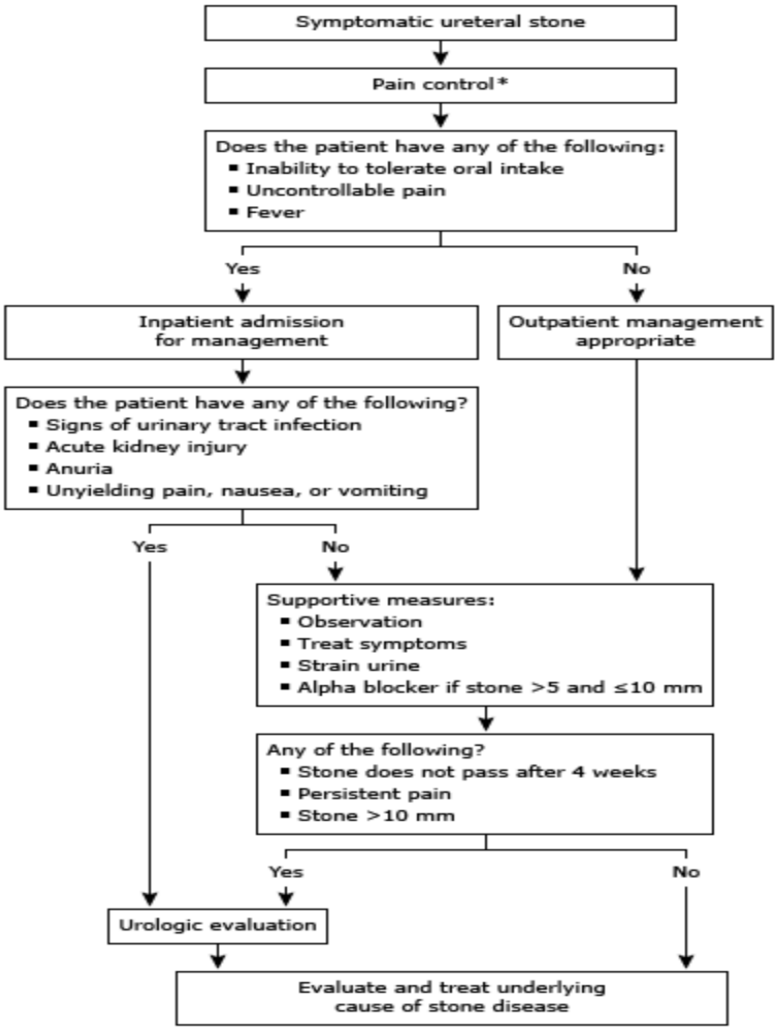
Ureterovesical junction

General Management

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- Non-contrast CT especially if first stone
- Pain management → NSAIDs or Opioids
- Assess size of stone
  a. < or = 5 mm it will pass spontaneously
  b. > 5 mm decreased chance of passage
    - Medications (like Alpha blockers) can facilitate passage
  c. > or = 10 mm passage is unlikely
- If concurrent UTI treat with abx
- Possible urology consult

General Management



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Treatment

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Calcium oxalate
- Low-sodium diet
- Thiazides

Calcium phosphate
- Low-sodium diet
- Thiazides
- Citrate

Struvite
- Eradication of underlying infection
- Surgical removal of stone

Uric acid
- Alkalinization of urine
- Allopurinol

Cystine
- Low-sodium diet
- Alkalinization of urine
- Chelating agents (e.g. tiopronin, penicillamine) if refractory

Stone Inhibitors

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- Citrate
  → Complexes with calcium and lowers urine calcium saturation
- Urea
- Magnesium
- Glycosaminoglycans
- Pyrophosphate

Reduce ionic activity and the formation of crystal complexes

Surgical Approaches

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EMERGENT INDICATIONS

a. Obstructing stones with UTI
b. Bilateral kidney stones with AKI
c. Unilateral obstruction with AKI in those with one functional kidney

ELECTIVE INDICATIONS

a. Stones > 10 mm
b. Uncomplicated distal stones < or = 10 mm that have not passed in 4 to 6 weeks with medical management
c. Symptomatic stones without any other etiology for pain
d. Pregnant patients who failed whom observation failed
e. Persistent kidney obstruction
f. Recurrent UTI related stones (Struvite)

Surgical Approaches

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EMERGENT

- Percutaneous nephrolithotomy (percutaneous lithotripsy)
- Ureteroscopy with ureteral stent

ELECTIVE

- Ureteroscopy
- Shock wave lithotripsy

Surgical Approaches

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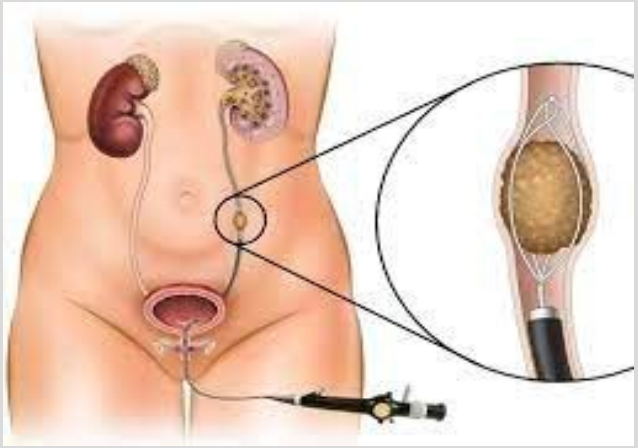
FOR STONES IN THE URETER

- Proximal and mid ureteral stones < or = to 10 mm → shock wave lithotripsy or ureteroscopy
- >10 mm → ureteroscopy
- Distal ureteral stones → ureteroscopy

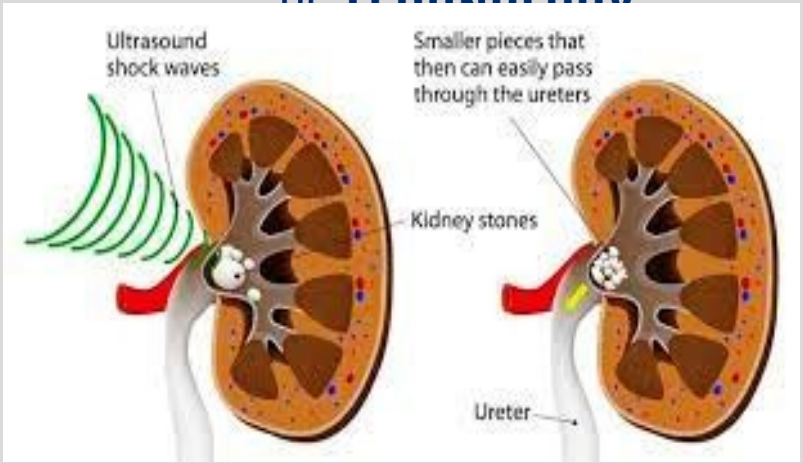
FOR STONES IN THE KIDNEY

- < or = 15 mm in upper pole, middle calyx, or pelvis → shock wave lithotripsy or ureteroscopy
- < or = 15 mm in lower pole → ureteroscopy or percutaneous nephrolithotomy
- > 15 mm in any region of kidney → percutaneous nephrolithotomy

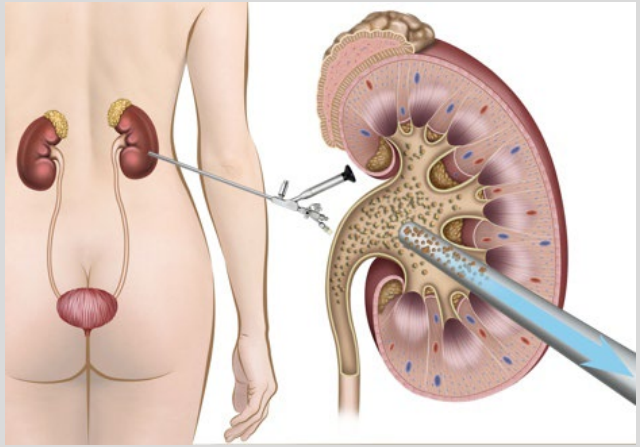
Surgical Approaches



Ureteroscopy



Shock wave lithotripsy



Percutaneous nephrolithotomy

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Review Questions

Question 1

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How does limiting sodium in the diet help prevent kidney stone formation?

Question 1 Answer

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Calcium reabsorption is tightly coupled to sodium reabsorption in the proximal tubule and loop of Henle (90% of calcium handling in the nephron). Decreased sodium intake leads to increased reabsorption of sodium and water, which increases passive calcium reabsorption. The resulting decreased urine calcium concentration reduces likelihood of stone formation.

Question 2

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What are the indications for a urology consult?

Question 2 Answer

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1. Superimposed UTI
2. AKI above expected
3. Anuria (complete obstruction)
4. Intractable pain, nausea, and vomiting
5. Stone not passed >4 weeks with medical management
6. >10 mm stone

Question 3

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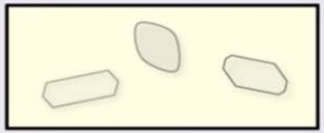
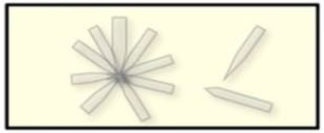
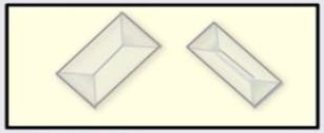
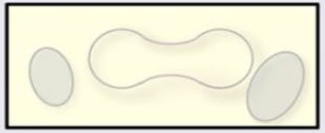
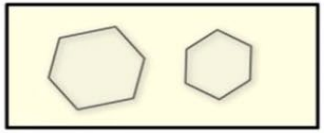
Why do people with Crohn’s Disease have increased incidence of kidney stones?

Question 3 Answer

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People with Crohn's disease are prone to malabsorption. This includes fats which are eliminated in through the stool. Excess fat in the stool binds to calcium releasing oxalate which is reabsorbed into the blood which is filtered out by the kidneys. This can lead to supersaturated concentrations of oxalate in the urine and stone formation.

Urine Microscopy Crystals

Crystals	Characteristics of Formation	Appearance	Diagnostic Utility
Uric Acid	Formation promoted by acidic urine		Seen in tumor lysis syndrome
Calcium phosphate	Formation promoted by alkaline urine		Not suggestive of any specific systemic disease
Magnesium ammonium phosphate (a.k.a. struvite or "triple phosphate")	Formation promoted by alkaline urine		Seen in UTIs by urease-producing organisms (e.g. *Proteus*, *Klebsiella*)
Calcium oxalate dihydrate	Formation is largely independent of urine pH		Not suggestive of any specific systemic disease
Calcium oxalate monohydrate	Formation is largely independent of urine pH		Seen in ethylene glycol ingestion
Cystine	Formation promoted by acidic urine		Diagnostic of cystinuria

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